

JAMES STROHM

FULL-STACK DEVELOPER

Belo Horizonte, Brazil · U.S. Citizen

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EXPERIENCE

CLUBE CERTO — Belo Horizonte, MG

2025 – Present

Full-Stack Developer

- Built and deployed 8+ RESTful microservices and 432+ Vue/React front-end components for a loyalty SaaS platform, architecting 281 database models across 32+ domain-driven feature modules.
- Integrated 5+ payment providers (Asaas, EFI, Vindi, GalaxPay) for PIX/Boleto transactions and external APIs for Firebase, telecom (IXC), and healthcare (Hinova).
- Developed 5 AWS SQS worker pipelines handling associate onboarding, lead management, WhatsApp messaging, and bulk Excel imports processing 50K+ rows.
- Implemented JWT auth with multi-role RBAC, Zod validation, real-time Socket.io, and an automated financial recovery system with multi-channel collection drivers.
- Translated Uber's H3 spatial indexing library from C to TypeScript and implemented multi-resolution hexagonal geolocation indexing with k-ring queries, achieving 2-10x faster establishment search performance.
- Built reusable front-end component systems including a 1,100-line dynamic AutoForm generator (16+ field types), enterprise DataTable, and real-time financial dashboards.

NODE.JS

TYPESCRIPT

VUE.JS

REACT

MYSQL

REDIS

AWS S3

AWS SQS

REST APIS

MICROSERVICES

GIT

COMFORTLY TRAVEL

2025 – 2026

Software Engineer (Back-end)

- Built and maintained Node.js/Express REST APIs deployed on AWS EC2 with Docker containerization for a travel booking platform.
- Optimized PostgreSQL queries and indexing strategies, reducing API latency by 25% and improving data access speed by 30%.
- Implemented CI/CD pipelines with GitHub Actions, reducing deployment errors by 80% and enabling reliable continuous delivery.

UPWARDTECH

2024 – 2026

Software Engineer (Consultant)

- Designed and built Flask + PostgreSQL CRM APIs for client projects, delivering custom business logic and data modeling.
- Automated client onboarding workflows, reducing manual setup time by 40% through programmatic provisioning.
- Improved production observability and incident response with AWS CloudWatch monitoring, logging, and alerting.

UNITED STATES AIR FORCE

2017 – 2022

Logistics Specialist

- Supported 5,000+ airlift missions across 24 countries; automated inventory tracking with Python on Linux, managing 33,000+ tons of cargo.

PROJECTS

AUTOMATED BILLING ENGINE Professional — Clube Certo

Automated multi-channel billing collection engine

- Built an automated collection system with rule-based triggers (D+3, D+7, etc.), CRON batch processing, and multi-channel delivery via pluggable email/WhatsApp/SMS drivers.
- Implemented email pixel tracking with unique UUIDs for open detection, rate-limited SMTP pooling (1 email/3s), and atomic transaction logging with retry on failure.
- Processed thousands of invoices hourly across companies with status filtering, amount ranges, weekend skipping, and a test mode for safe staging rollout.

NODE.JS

MYSQL

CRON

SMTP

WHATSAPP API

STATE MACHINE

H3 GEOLOCATION SYSTEM Professional — Clube Certo

Hexagonal spatial indexing for high-performance geolocation queries

- Translated Uber's H3 spatial indexing library from C to JavaScript/TypeScript and implemented multi-resolution hexagonal indexing (~1.2km, ~460m, ~174m) with k-ring expansion queries for 2-10x faster establishment searches.
- Built Sequelize model hooks for automatic H3 cell calculation on address create/update, dual query builders (closest address vs. full multi-address), and migration scripts for zero-downtime backfill of existing records.

C TYPESCRIPT H3-JS MYSQL SEQUELIZE NODE.JS

HERO [VIEW SITE](#) Professional — Clube Certo

Enterprise admin dashboard with 432 components across 42 feature modules

- Built a 432-component Vue 3 + TypeScript admin dashboard with TanStack Query for data fetching, shadcn-vue component system, and dark mode support.
- Developed a dynamic AutoForm generator (1,100+ lines, 16+ field types) and enterprise DataTable (850+ lines) with pagination, sorting, sticky headers, and global filtering.
- Implemented role-based permission controls, financial dashboards with real-time metrics, and a feature-driven modular architecture with isolated routing per module.

VUE 3 TYPESCRIPT TANSTACK QUERY TAILWIND CSS SHADCN-VUE VITE

A.L.F.R.E.D [GITHUB](#) Personal — Voice-Enabled AI Assistant

All-Knowing Logical Facilitator for Reasoned Execution of Duties

- Engineered a voice-enabled desktop assistant with multi-provider LLM fallback (OpenAI, Claude), ElevenLabs TTS, and a PySide6 GUI featuring real-time waveform visualization and system dashboards.
- Implemented semantic memory via ChromaDB vector storage with OpenAI embeddings, plus Google Calendar, weather, file search, and system automation integrations.

PYTHON PYSIDE6 OPENAI API CHROMADB ELEVENLABS

CHESS ENGINE + AI [GITHUB](#) Personal — C++17 / SDL2

Chess engine with AI opponent using minimax search, alpha-beta pruning, and SDL2 GUI

- Built a chess AI with minimax + alpha-beta pruning, quiescence search to resolve capture sequences, and a killer move heuristic for faster pruning at depth 5.
- Implemented positional evaluation with king safety, pawn structure analysis (doubled/isolated/passed pawns), bishop pair bonuses, and piece-square tables.
- Developed an SDL2 graphical interface with click-to-move interaction, move highlighting, undo, live scoreboard, and real-time status messages; compiled to WebAssembly via Emscripten for in-browser play.

C++17 SDL2 CMAKE OOP DATA STRUCTURES MINIMAX ALPHA-BETA ALGORITHMS WEBASSEMBLY EMSCRIPTEN

EDUCATION

ECPI UNIVERSITY

B.S. Information Technology (Software Development)

GPA: 4.0

2024 – 2026

CODING TEMPLE

Certificate, Software Engineering

2024

TECHNICAL SKILLS

BACKEND	Python, PHP, Node.js, TypeScript, JavaScript, Express, Flask, Django, REST APIs, Microservices, JWT, OAuth2, Socket.io, Zod, h3-js, Firebase, Stripe
FRONTEND	React, React Native, Vue.js, Python, Tailwind CSS, HTML5, CSS3
DATABASE	MySQL, PostgreSQL, MongoDB, Sequelize, Redis
CLOUD & DEVOPS	AWS EC2, S3, SQS, CloudWatch, Docker, Git, GitHub, GitHub Actions, CI/CD, Linux
TESTING & METHODS	Jest, Mocha, Agile, Scrum, Kanban, Code Review
OTHER	C, C++, SDL2, CMake

LANGUAGES

English — Native Portuguese — Native Spanish — Professional